

Hotel Management System

Group Project Module Allocation & System Overview

1. System Overview

The Hotel Management System (HMS) is designed to manage hotel operations such as room reservations, user management, food ordering, inventory handling, supplier management, and payments.

The system supports three main user roles:

- Admin
- Customer
- Supplier

Each user accesses the system using a unique username and password.

2. Special System Rules & Features

User Roles & Access Control

1. Admin Registration Restriction

- Not everyone can register as an Admin.
- Admins are allocated by hotel management as a job role.
- Admin accounts are created and controlled by the hotel.
- Customers and suppliers cannot self register as admins.

2. User Authentication

- Every Customer, Supplier, and Admin has:
 - A unique User ID
 - A username and password
- Login is required to access system features.

Customer Features

3. Customers fill a registration form with personal details.

4. Customers can:

- View available rooms
- Reserve rooms
- View menu and place food orders

5. Each customer has:

- Unique User ID
- Reservation history

Room & Reservation Management

6. Each reservation has a unique Reservation ID.
7. Room status is updated automatically based on reservations.
8. Admin manages room availability (Available / Reserved / Occupied).

Food Ordering & Billing

9. Customers can:
 - View menu
 - Order food
10. Final bill includes:
 - Room charges
 - Food order total
11. A final bill is generated for each room.

Supplier & Inventory

12. Suppliers can:
 - View supply forms
 - Add and update supply items
 - View their past supply records
13. Admin can:
 - View inventory records
 - Update stock
 - View inventory history Reports & History 1
14. Admin can view:
 - Reservation history
 - Inventory history
 - Payment history
 - Early payment records

3.Task Allocation by Group Members

I.A.D.T.E.Medhawana :AS20240556

Module: Authentication & Welcome Page

- Create welcome page
- Login system
- Registration form
- Store user details in database
- Implement role based navigation

M.A.B.Vihangi :AS20240449

Module: Reservation Management

- View rooms and facilities
- Show room prices
- Reserve rooms using User ID
- Generate Reservation ID
- Store reservation details in database
- Customer can view their own reservation history.

B.M.Herath :AS20240612 & R.P.K.Wijesinghe :AS20240599

Module: Admin Management

- Add, update, delete rooms
- Manage room availability
- Inventory management (Update,add stock)
- View available stock
- View reports:
 - Reservations
 - Payments
 - Inventory history

R.A.D.Mihiwanthi :AS20240452

Module: Food Ordering

- View menu
- Place food orders
- Calculate total food cost
- View order details (food ID, price, quantity)
- Track order delivery time

B.M.N.Y.Bandaranayaka :AS20240412

Module: Supplier Inventory

- Supplier profile management

- Add/update supply items
- Increase inventory quantity
- View past supply records

U.D.G.I.Erandi :AS20240595

Module: Payment System

- Generate Payment ID
- Calculate total bill (room + food)
- Payment status (Paid / Pending)
- Select payment method
- Generate receipt
- Save payment details in database